Physics 710-712 April 16, 2010

Problem Set 18

Problem 14.4.3: Schrödinger's equation is

$$i\hbar\frac{d}{dt}|\psi\rangle = -\gamma\vec{S}\cdot\vec{B}|\psi\rangle, \qquad \text{with} \quad \vec{B} = B(\cos\omega t\,\hat{\imath} - \sin\omega t\,\hat{\jmath}) + B_0\hat{k}. \tag{1}$$

Define $\omega_0 \equiv \gamma B_0$ and $\omega_r \equiv \gamma\sqrt{B^2 + (B_0 - \frac{\omega}{\gamma})^2} = \sqrt{\gamma^2 B^2 + (\omega-\omega_0)^2}$. Also, define

$$|\psi_r\rangle \equiv e^{-i\omega t S_z/\hbar}|\psi\rangle,$$

in terms of which (1) becomes $i\hbar\frac{d}{dt}[e^{i\omega t S_z/\hbar}|\psi_r\rangle] = -\gamma\vec{S}\cdot\vec{B}e^{i\omega t S_z/\hbar}|\psi_r\rangle$. Expand out the derivative on the left-hand side, multiply through by $e^{-i\omega t S_z/\hbar}$, and use the fact that $[S_z, e^{i\omega t S_z/\hbar}] = 0$ to get

$$\begin{aligned}
-\omega S_z|\psi_r\rangle + i\hbar\frac{d}{dt}|\psi_r\rangle &= -\gamma e^{-i\omega t S_z/\hbar}\vec{S}\cdot\vec{B}e^{+i\omega t S_z/\hbar}|\psi_r\rangle \\
&= -\gamma\frac{\hbar}{2}B_0\sigma_z|\psi_r\rangle - \gamma B\frac{\hbar}{2}e^{-i\omega t\sigma_z/2}\left(\cos\omega t\,\sigma_x - \sin\omega t\,\sigma_y\right)e^{i\omega t\sigma_z/2}|\psi_r\rangle,
\end{aligned}$$

which implies

$$\begin{aligned}
2i\frac{d}{dt}|\psi_r\rangle &= (\omega-\omega_0)\sigma_z|\psi_r\rangle \\
&\quad - \gamma B\left[\cos(\omega t/2) - i\sigma_z\sin(\omega t/2)\right]\left[\sigma_x\cos(\omega t) - \sigma_y\sin(\omega t)\right]\left[\cos(\omega t/2) + i\sigma_z\sin(\omega t/2)\right]
\end{aligned} \tag{2}$$

But

$$\begin{aligned}
&\left[\cos(\omega t/2) - i\sigma_z\sin(\omega t/2)\right]\left[\sigma_x\cos(\omega t) - \sigma_y\sin(\omega t)\right]\left[\cos(\omega t/2) + i\sigma_z\sin(\omega t/2)\right] \\
&= \left[\sigma_x c_2 c - i\sigma_z\sigma_x s_2 c - \sigma_y c_2 s + i\sigma_z\sigma_y s_2 s\right]\left[c_2 + i\sigma_z s_2\right] \\
&= \left[\sigma_x c_2 c - i(i\sigma_y)s_2 c - \sigma_y c_2 s + i(-i\sigma_x)s_2 s\right]\left[c_2 + i\sigma_z s_2\right] \\
&= \left[\sigma_x(c_2 c + s_2 s) + \sigma_y(s_2 c - c_2 s)\right]\left[c_2 + i\sigma_z s_2\right] \\
&= \left[\sigma_x c_2 - \sigma_y s_2\right]\left[c_2 + i\sigma_z s_2\right] \\
&= \sigma_x c_2^2 + i\sigma_x\sigma_z c_2 s_2 - \sigma_y c_2 s_2 - i\sigma_y\sigma_z s_2^2 \\
&= \sigma_x c_2^2 + i(-i\sigma_y)c_2 s_2 - \sigma_y c_2 s_2 - i(i\sigma_x)s_2^2 \\
&= \sigma_x(c_2^2 + s_2^2) + \sigma_y(c_2 s_2 - c_2 s_2) \\
&= \sigma_x,
\end{aligned}$$

where I've introduced the shorthands $c \equiv \cos\omega t$, $s \equiv \sin\omega t$, $c_2 \equiv \cos(\omega t/2)$, $s_2 \equiv \sin(\omega t/2)$, and I've used trig identities and the σ-matrix algebra $\sigma_x\sigma_y = i\sigma_z$ and cyclic permutations. This then implies that (2) becomes

$$i\frac{d}{dt}|\psi_r\rangle = \widehat{H}|\psi_r\rangle \qquad \text{with} \quad \widehat{H} = \frac{\omega-\omega_0}{2}\sigma_z - \frac{\gamma B}{2}\sigma_x,$$

which has the form of the Schrödinger equation, but now with a time-independent Hamiltonian, $\hbar\widehat{H}$. So its solution is $|\psi_r(t)\rangle = U_r(t)|\psi_r(0)\rangle$ with $U_r(t) = e^{-i\widehat{H}t}$. Therefore

$$|\psi(t)\rangle = e^{i\omega t S_z/\hbar}|\psi_r(t)\rangle = e^{i\omega t S_z/\hbar}e^{-i\widehat{H}t}|\psi_r(0)\rangle = e^{i\omega t S_z/\hbar}e^{-i\widehat{H}t}e^{-i\omega t S_z/\hbar}|\psi(0)\rangle.$$

Now, in the S_z-basis, $|\psi(0)\rangle = \binom{1}{0}$. Also, write $e^{-i\widehat{H}t} = e^{-i\vec{\theta}\cdot\vec{S}/\hbar}$ with $\vec{\theta} = -\gamma B t\widehat{\imath} + (\omega - \omega_0)t\widehat{k}$. So, $e^{-i\widehat{H}t} = \cos(\theta/2) - i(\widehat{\theta}\cdot\vec{\sigma})\sin(\theta/2)$ where

$$\begin{aligned}
\theta &= \sqrt{(\gamma B)^2 + (\omega-\omega_0)^2}\; t \;\equiv\; \omega_r t, \\
\widehat{\theta} &= \frac{-\gamma B\widehat{\imath} + (\omega-\omega_0)\widehat{k}}{\sqrt{(\gamma B)^2 + (\omega-\omega_0)^2}} \;=\; -\frac{\gamma B}{\omega_r}\widehat{\imath} + \frac{\omega-\omega_0}{\omega_r}\widehat{k}.
\end{aligned}$$

So we get

$$e^{-i\widehat{H}t} = \cos\left(\frac{\omega_r t}{2}\right) - i\sin\left(\frac{\omega_r t}{2}\right)\left[\frac{\omega-\omega_0}{\omega_r}\sigma_z - \frac{\gamma B}{\omega_r}\sigma_x\right].$$

Thus, we have

$$\begin{aligned}
|\psi(t)\rangle &= \left[\cos\tfrac{\omega t}{2} + i\sigma_z\sin\tfrac{\omega t}{2}\right]\left[\cos\tfrac{\omega_r t}{2} - i\sin\tfrac{\omega_r t}{2}\left(\tfrac{\omega-\omega_0}{\omega_r}\sigma_z - \tfrac{\gamma B}{\omega_r}\sigma_x\right)\right]\left[\cos\tfrac{\omega t}{2} - i\sigma_z\sin\tfrac{\omega t}{2}\right]\begin{pmatrix}1\\0\end{pmatrix} \\
&= \left[\cos\tfrac{\omega t}{2} + i\sigma_z\sin\tfrac{\omega t}{2}\right]\left[\cos\tfrac{\omega_r t}{2} - i\sin\tfrac{\omega_r t}{2}\left(\tfrac{\omega-\omega_0}{\omega_r}\sigma_z - \tfrac{\gamma B}{\omega_r}\sigma_x\right)\right]\begin{pmatrix}e^{-i\omega t/2}\\0\end{pmatrix} \\
&= \begin{pmatrix}e^{i\omega t/2} & 0\\ 0 & e^{-i\omega t/2}\end{pmatrix}\begin{pmatrix}\left[\cos\tfrac{\omega_r t}{2} - i\tfrac{\omega-\omega_0}{\omega_r}\sin\tfrac{\omega_r t}{2}\right]e^{-i\omega t/2} \\ i\tfrac{\gamma B}{\omega_r}\sin\tfrac{\omega_r t}{2}\; e^{-i\omega t/2}\end{pmatrix} \\
&= \begin{pmatrix}\cos\tfrac{\omega_r t}{2} - i\tfrac{\omega-\omega_0}{\omega_r}\sin\tfrac{\omega_r t}{2} \\ i\tfrac{\gamma B}{\omega_r}\sin\tfrac{\omega_r t}{2}\; e^{-i\omega t}\end{pmatrix}.
\end{aligned}$$

(Note: I disagree with the result in the book by an overall factor of $e^{i\omega t/2}$, which, however, is unobservable.)

When $\omega = \omega_0$ we have $\omega_r = \gamma B$, and so

$$|\psi(t)\rangle = \begin{pmatrix}\cos\tfrac{\omega_r t}{2} \\ i\sin\tfrac{\omega_r t}{2}\; e^{-i\omega_0 t}\end{pmatrix} = e^{-i\frac{\omega_r}{2}t + i\frac{\pi}{4}}\begin{pmatrix}\cos\tfrac{\omega_r t}{2}\; e^{i(\omega_0 t - \pi/2)/2} \\ i\sin\tfrac{\omega_r t}{2}\; e^{-i(\omega_0 t-\pi/2)/2}\end{pmatrix}.$$

Comparing this to

$$|\widehat{n}+\rangle = \begin{pmatrix}\cos\tfrac{\theta}{2}\; e^{-i\phi/2} \\ i\sin\tfrac{\theta}{2}\; e^{+i\phi/2}\end{pmatrix},$$

we see that up to a phase, $|\psi(t)\rangle$ is the same with

$$\theta = \omega_r t, \qquad \text{and} \qquad \phi = \frac{\pi}{2} - \omega_0 t.$$

Thus, in addition to precessing around the z-axis with angular velocity ω_0, it is also flipping from $+z$ to $-z$ with angular velocity ω_r.

Finally, we calculate $\langle\mu_z(t)\rangle$:

$$\begin{aligned}
\langle\mu_z(t)\rangle &= \langle\psi(t)|\gamma S_z|\psi(t)\rangle = \frac{\gamma\hbar}{2}\langle\psi(t)|\sigma_z|\psi(t)\rangle \\
&= \frac{\gamma\hbar}{2}\langle\psi(t)|\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\begin{pmatrix}\cos\frac{\omega_r t}{2} - i\frac{\omega-\omega_0}{\omega_r}\sin\frac{\omega_r t}{2}\\ i\frac{\gamma B}{\omega_r}\sin\frac{\omega_r t}{2}\, e^{-i\omega t}\end{pmatrix} \\
&= \frac{\gamma\hbar}{2}\begin{pmatrix}\cos\frac{\omega_r t}{2} + i\frac{\omega-\omega_0}{\omega_r}\sin\frac{\omega_r t}{2} & -i\frac{\gamma B}{\omega_r}\sin\frac{\omega_r t}{2}\, e^{+i\omega t}\end{pmatrix}\begin{pmatrix}\cos\frac{\omega_r t}{2} - i\frac{\omega-\omega_0}{\omega_r}\sin\frac{\omega_r t}{2}\\ -i\frac{\gamma B}{\omega_r}\sin\frac{\omega_r t}{2}\, e^{-i\omega t}\end{pmatrix} \\
&= \frac{\gamma\hbar}{2}\left[\cos^2\frac{\omega_r t}{2} + \left(\frac{\omega-\omega_0}{\omega_r}\right)^2\sin^2\frac{\omega_r t}{2} - \frac{\gamma^2B^2}{\omega_r^2}\sin^2\frac{\omega_r t}{2}\right] \\
&= \frac{\gamma\hbar}{2}\left[\cos^2\frac{\omega_r t}{2} + \frac{(\omega-\omega_0)^2-\gamma^2B^2}{\omega_r^2}\sin^2\frac{\omega_r t}{2}\right] \\
&= \frac{\gamma\hbar}{2\omega_r^2}\left([(\omega-\omega_0)^2+\gamma^2B^2]\cos^2\frac{\omega_r t}{2} + [(\omega-\omega_0)^2-\gamma^2B^2]\sin^2\frac{\omega_r t}{2}\right) \\
&= \frac{\gamma\hbar}{2\omega_r^2}\left[(\omega-\omega_0)^2+\gamma^2B^2\cos(\omega_r t)\right].
\end{aligned}$$

Problem 14.4.4: Rotate the whole problem by $\pi/2$ so $\vec{B} = B\widehat{k}$ and $|\psi(0)\rangle$ is the $S_x = -\hbar/2$ eigenstate. Then, in the S_z-basis, $|\psi(0)\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix}$, and states evolve by $U(t) = \begin{pmatrix}e^{i\omega_0 t/2} & 0\\ 0 & e^{-i\omega_0 t/2}\end{pmatrix}$, where $\omega_0 \equiv \gamma B$. So

$$|\psi(t)\rangle = U(t)|\psi(0)\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}e^{i\omega_0 t/2}\\ -e^{-i\omega_0 t/2}\end{pmatrix}.$$

The question is, what is t when $|\psi(t)\rangle$ is the $+(\hbar/2)$ S_x eigenstate? In other words, when is

$$|\psi(t)\rangle \propto \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix} \quad ?$$

This implies we need $e^{i\omega_0 t/2} = -e^{-i\omega_0 t/2}$ which implies $\omega_0 t = (2n+1)\pi$ for n an integer. So the shortest time is when $n=0$, or $t = \pi/\omega_0 = \pi/(\gamma B)$.

For an electron spin, $\gamma = e/(mc) \approx 1.5\times 10^7\,\mathrm{G}^{-1}\mathrm{s}^{-1}$, so $t = \pi/(\gamma B) \approx 2\times 10^{-9}$s.

Problem 14.5.4: For spin 1, $m \in \{1, 0, -1\}$ which we will associate with the orthonormal basis of states $\{|1\rangle, |0\rangle, |-1\rangle\}$. Then the uppermost beam of the first Stern-Gerlach (SG) apparatus (*i.e.*, the with magnetic field along the z-axis) will be in the pure $S_z = +\hbar$ state $|1\rangle$. The second SG (z'-axis) apparatus then separates the beam according to $\widehat{n}'\cdot\vec{S} = \{+\hbar, 0, -\hbar\}$ eigenstates, where

$$\widehat{n}' = \cos\theta\,\widehat{k} + \sin\theta\,\widehat{\imath}, \quad \text{implying} \quad \widehat{n}'\cdot\vec{S} = \cos\theta\, S_z + \sin\theta\, S_x = \cos\theta\, S_z + \tfrac{1}{2}\sin\theta\,(S_+ + S_-). \tag{3}$$

The uppermost beam is then in the pure $\widehat{n}'\cdot\vec{S} = +\hbar$ eigenstate $|1'\rangle$. Therefore, the probability of observing the $|1'\rangle$ state is

$$\text{Prob} = |\langle 1'|1\rangle|^2. \tag{4}$$

So we need to compute $|1'\rangle$ in terms of the $\{|1\rangle, |0\rangle, |-1\rangle\}$ basis. Say $|1'\rangle = a|1\rangle + b|0\rangle + c|-1\rangle$. Since, by definition, $(\widehat{n}'\cdot\vec{S})|1'\rangle = +\hbar|1'\rangle$, we have

$$\hbar\,(a|1\rangle + b|0\rangle + c|-1\rangle) = (\widehat{n}'\cdot\vec{S})\,(a|1\rangle + b|0\rangle + c|-1\rangle)\,. \tag{5}$$

Using (3) gives

$$\begin{aligned}
\widehat{n}'\cdot\vec{S}|m\rangle &= \cos\theta\, S_z|m\rangle + \tfrac{1}{2}\sin\theta(S_+ + S_-)|m\rangle \\
&= \hbar m\cos\theta|m\rangle + \tfrac{\hbar}{2}\sin\theta\left(\sqrt{(1-m)(2+m)}|m+1\rangle + \sqrt{(1+m)(2-m)}|m-1\rangle\right),
\end{aligned}$$

implying

$$\begin{aligned}
\widehat{n}'\cdot\vec{S}|1\rangle &= \hbar\cos\theta|1\rangle + \tfrac{\hbar}{\sqrt{2}}\sin\theta|0\rangle, \\
\widehat{n}'\cdot\vec{S}|0\rangle &= \tfrac{\hbar}{\sqrt{2}}\sin\theta|1\rangle + \tfrac{\hbar}{\sqrt{2}}\sin\theta|-1\rangle, \\
\widehat{n}'\cdot\vec{S}|-1\rangle &= -\hbar\cos\theta|-1\rangle + \tfrac{\hbar}{\sqrt{2}}\sin\theta|0\rangle.
\end{aligned}$$

Plugging these into (5) gives

$$a|1\rangle + b|0\rangle + c|-1\rangle = \left(a\cos\theta + \tfrac{b}{\sqrt{2}}\sin\theta\right)|1\rangle + \left(\tfrac{a}{\sqrt{2}}\sin\theta + \tfrac{c}{\sqrt{2}}\sin\theta\right)|0\rangle + \left(\tfrac{b}{\sqrt{2}}\sin\theta - c\cos\theta +\right)|-1\rangle.$$

Solving for a, b, and c and imposing that $|1'\rangle$ be normalized (*i.e.*, that $|a|^2 + |b|^2 + |c|^2 = 1$) gives

$$a = \frac{1+\cos\theta}{2}, \qquad b = \frac{\sin\theta}{\sqrt{2}}, \qquad c = \frac{1-\cos\theta}{2}.$$

Therefore $\langle 1'|1\rangle = a = (1+\cos\theta)/2$, and so by (4)

$$\text{Prob} = |\langle 1'|1\rangle|^2 = \left(\frac{1+\cos\theta}{2}\right)^2.$$